SIMATS Engineering • CSE • CSA0802 — Python Programming • 29-07-2026 • Category: CO2_AT3_Coding Challenge • Student Submissions Report

Student 5 **DANIMIZA V** 192421363RC 4 / 4 submitted

Q1. String Character Frequency Counter

CODE

```
s = input().lower()
count = {}
for ch in s:
    if ch ≠ " ":
        if ch in count:
            count[ch] += 1
        else:
            count[ch] = 1
for ch in sorted(count):
    print(ch,"→",count[ch])
```

INPUT

```
Hello World
```

OUTPUT

```
d -> 1
e -> 1
h -> 1
l -> 3
o -> 2
r -> 1
w -> 1
```

Q2. Vowel and Consonant Classifier with Statistics

CODE

```
s = input().lower()
count = {}
for ch in s:
    if ch ≠ " ":
        if ch in count:
            count[ch] += 1
        else:
            count[ch] = 1
for ch in sorted(count):
    print(ch,"→",count[ch])
```

INPUT

```
Hello World
```

OUTPUT

```
d -> 1
e -> 1
h -> 1
l -> 3
o -> 2
r -> 1
w -> 1
```

Q3. String Compression and Run-Length Encoding

CODE

```
s = input()
c =""
i = 0
while i < len(s):
    count = 1
    while i < len(s)-1 and s[i] == s[i+1]:
        count += 1
        i += 1
    c += s[i] + (str(count) if count > 1 else "")
    i += 1
if len(c) < len(s):
    print("compressed:",c)
else:
    print("original:",s)
```

INPUT

```
aabbbccdddee
```

OUTPUT

```
compressed: a2b3c2d3e2
```

Q4. Advanced Text Statistics Analyser

CODE

```
form collections import counter
import re
t = input()
w = re.findall(r'\b\w+\b',t.lower())
s = re.split(r'[.!?]+',t.strip())
stop = {"the","is","a","an","in","of","and","to","for"}
f = [i for i in w if i not in stop]
c = counter(f)
print("total words:",len(w))
print("sentences:",len[i for i in s if i])
print("paragraphs:",1)
print("avg word length:",round(sum(map(len,w))/len(w),2))
print("avg sentence length:",round(len(w)/max(1,len([i for i in s if i])),2))
print("top 5 words:",c.most_common(5))
```

INPUT

```
python is a powerful programming language
```

OUTPUT

```
  File "C:\Windows\TEMP\sandbox_192421363RC_6a625da88fabf3.47163693\main.py", line 1
    form collections import counter
         ^^^^^^^^^^^
SyntaxError: invalid syntax
```

Student 3 **DANIMIZA V** 192421363RC 20 / 20 submitted

Q1. Void Function (No Return Value)

CODE

```
def welcome():
    print("welcome to python programming")
welcome()
```

OUTPUT

```
(no output)
```

Q2. Void Function (No Return Value)

CODE

```
def multiplication_table(n):
    for i in range(1,11):
        print(n,"*",i,"=",n*i)
num = int(input("enter a number:"))
multiplication_table(num)
```

INPUT

```
5
```

OUTPUT

```
(no output)
```

Q3. Void Function (No Return Value)

CODE

```
def print_even(n):
    for i in range(1, n+1):
        if i % 2 == 0:
            print(i)
num = int(input("enter n:"))
print_even(num)
```

INPUT

```
10
```

OUTPUT

```
(no output)
```

Q4. Fruitful Function (Returns One Value)

CODE

```
def count_vowels(text):
    count = 0
    vowels = "aeiouAEIOU"
    for ch in text:
        if ch in vowels:
            count += 1
    return count
string = input("enter a string:")
result = count_vowels(string)
print("\nnumber of vowels:",result)
```

INPUT

```
Hello World
```

OUTPUT

```
(no output)
```

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Q5. Fruitful Function (Returns One Value)

CODE

```
def largest_element(numbers):
    largest = numbers[0]
    for num in numbers:
        if num > largest:
            largest = num
    return largest
n = int(input("enter the number of elements:"))
numbers = []
for i in range(n):
    num = int(input("enter element:"))
    numbers.append(num)
result = largest_element(numbers)
print("\nlargest element:",result)
```

INPUT

```
5
12
45
7
89
23
```

OUTPUT

```
(no output)
```

Q6. Fruitful Function (Returns One Value)

CODE

```
def is_prime(n):
    if n <= 1:
        return "not prime"
    for i in range(2, n):
        if n % i == 0:
            return "not prime"
    return "prime"
num = int(input("enter a number:"))
result = is_prime(num)
print(result)
```

INPUT

```
7
```

OUTPUT

```
(no output)
```

Q7. Function Returning Two or More Values

CODE

```
def student_result(mark1,mark2,mark3):
    total = mark1 + mark2 + mark3
    average = total / 3
    if average ≥ 90:
        grade = "A"
    elif average ≥ 75:
        grade = "B"
    elif average ≥ 50:
        grade = "C"
    else:
        grade = "fail"
    return total, average, grade
m1 = int(input("enter mark 1:"))
m2 = int(input("enter mark 2:"))
m3 = int(input("enter mark 3:"))
total, average, grade = student_result(m1, m2, m3)
print("total:",total)
print("average:",average)
print("grade:",grade)
```

INPUT

```
85
90
78
```

OUTPUT

```
(no output)
```

Q8. Function Returning Two or More Values

CODE

```
def circle(radius):
    pi = 3.14
    area = pi * radius * radius
    circumference = 2 * pi * radius
    return area, circumference
r = float(input("enter the radius:"))
area, circumference = circle(r)
print("\narea =",area)
print("\ncircumference=",circumference)
```

INPUT

```
5
```

OUTPUT

```
(no output)
```

Q9. Function Returning Two or More Values

CODE

```
def calculate(num):
    square = num ** 2
    cube = num ** 3
    square_root = num ** 0.5
    return square, cube, square_root
n = float(input("enter a number:"))
square, cube, square_root = calculate(n)
print("\nsquare =",square)
print("\ncube =",cube)
print("\nsquare_root =",square_root)
```

INPUT

```
4
```

OUTPUT

```
(no output)
```

Q10. Keyword Arguments

CODE

```
def student(name, age, department):
    print("name:",name)
    print("\nage:",age)
    print("\ndepartment:",department)
student(name="Danimiza",age="19",department="computer science")
```

OUTPUT

```
(no output)
```

Q11. Keyword Arguments

CODE

```
def employee(empid, name, salary):
    print("employee ID:",empid)
    print("\nemployee name:",name)
    print("\nemployee salary:",salary)
employee(salary=50000,name="john",empid=101)
```

OUTPUT

```
(no output)
```

Q12. Keyword Arguments

CODE

```
def book(title, author, price):
    print("title:",title)
    print("\nauthor:",author)
    print("\nprice:",price)
book(price=4500,author="jeon jungkook",title="undesired love")
```

OUTPUT

```
(no output)
```

Q13. Variable-Length Arguments (*args)

CODE

```
def find_sum(*args):
    return sum(args)
result = find_sum(10,20,30,40,50)
print("sum=",result)
```

OUTPUT

```
(no output)
```

Q14. Variable-Length Arguments (*args)

CODE

```
def average_marks(*marks):
    return sum(marks) / len(marks)
avg = average_marks(85,90,78,88,95)
print("average marks =",avg)
```

OUTPUT

```
(no output)
```

Q15. Variable-Length Arguments (*args)

CODE

```
def print_strings(*args):
    for string in args:
        print(string)
print_strings("python","java","c++","javascript","HTML")
```

OUTPUT

```
(no output)
```

Q16. Variable-Length Arguments (*args)

CODE

```
def largest_value(*args):
    return max(args)
result = largest_value(10,45,78,23,99,56)
print("largest value =",result)
```

OUTPUT

```
(no output)
```

Q17. Recursive Functions

CODE

```
def fibonacci(n):
    if n ≤ 1:
        return n
    return fibonacci(n-1) + fibonacci(n-2)
n = int(input("enter the number of terms:"))
print("\nfibonacci series:")
for i in range(n):
    print(fibonacci(i), end=" ")
```

INPUT

```
7
```

OUTPUT

```
(no output)
```

Q18. Recursive Functions

CODE

```
def sum_digits(n):
    if n == 0:
        return 0
    return (n % 10) + sum_digits(n // 10)
num = int(input())
print(sum_digits(num))
```

INPUT

```
1234
```

OUTPUT

```
(no output)
```

Q19. Recursive Functions

CODE

```
def rev(s):
    if len(s) == 0:
        return s
    return rev(s[1:]) + s[0]
text = input()
print(rev(text))
```

INPUT

```
hello
```

OUTPUT

```
(no output)
```

Q20. Recursive Functions

CODE

```
def power(x, y):
    if y == 0:
        return 1
    return x * power(x, y - 1)
base = int(input())
exp = int(input())
print(power(base, exp))
```

INPUT

```
2
3
```

OUTPUT

```
(no output)
```